

Software Requirement Specification (SRS)

Project Name: College Placement Management Portal



Revision History

Version	Date of Release	Author	Description of Changes	Signature
1.0	05/06/2026	PGCP-AC-026	Initial draft of SRS document	
1.1	10/06/2026	PGCP-AC-026	Added detailed functional requirements, non-functional requirements, acceptance criteria, dependencies, and revised revision history	

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1.0 Project Code

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2.0 Title of the project

College Placement Management Portal

3.0 Introduction

3.1 Purpose

The College Placement Management Portal is a web-based application designed to automate and streamline placement-related activities for colleges, universities, and educational institutions.

The system provides a centralized platform for managing students, recruiters, placement drives, applications, notifications, and placement reports. It eliminates manual processes, improves communication among stakeholders, and ensures transparency throughout the recruitment lifecycle.

3.2 Document Conventions

This document follows standard SRS formatting conventions. Requirements are written using **SHALL** for mandatory features and **SHOULD** for optional ones. Headings are hierarchical and all technical terms are defined in Section 3.6.

3.3 Intended Audience

This document is intended for:

- Project Guide
- Development Team
- Testing Team
- System Administrators
- Placement Administrators
- Recruiters
- Future Maintenance Team
- Project Evaluators

3.4 Project Scope

The College Placement Management Portal is designed to serve colleges, universities, and educational institutions by providing a centralized platform for managing placement activities involving students, placement administrators, recruiters, and system administrators.

The system facilitates:

- Student account management
- Student profile and resume management
- Recruiter and company management
- Placement drive management
- Eligibility verification
- Job application tracking
- Notifications and announcements
- Placement reporting and analytics

The application can be deployed within institutional infrastructure and accessed through standard web browsers.

3.5 References

- Project Workflow Document
- C-DAC SRS Template
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications
- React Documentation
- Spring Boot Official Documentation
- MySQL Documentation
- REST API Design Guidelines
- REST JWT Authentication

3.6 Definitions and Acronyms

Term	Description
SRS	Software Requirements Specification
JWT	JSON Web Token
API	Application Programming Interface
UI	User Interface
DB	Database
HR	Human Resources
CGPA	Cumulative Grade Point Average
Admin	Placement Administrator

4.0 Overall Description

4.1 Project Perspective

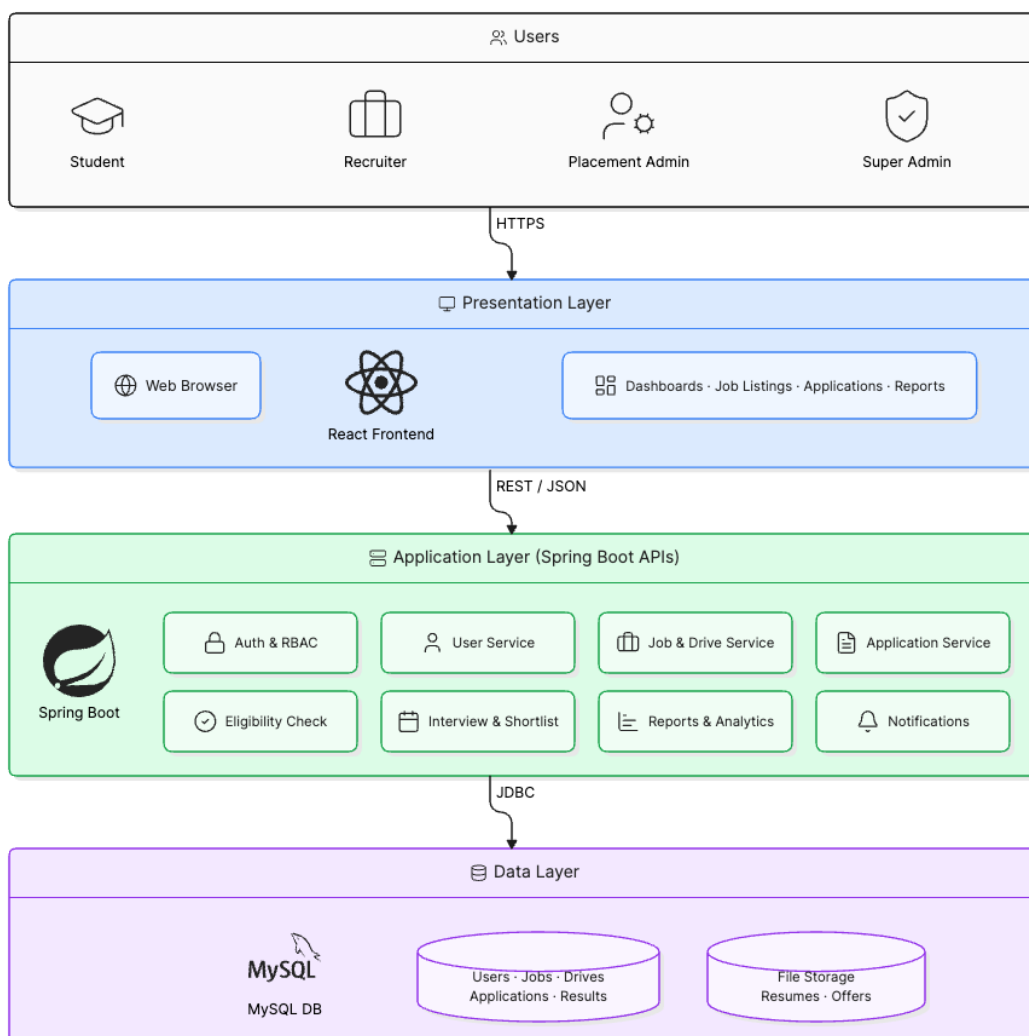
Traditional placement management systems rely heavily on manual processes that are time-consuming and prone to errors.

The proposed system acts as a centralized platform connecting:

- Students
- Placement Admins
- Recruiters
- Super Administrators

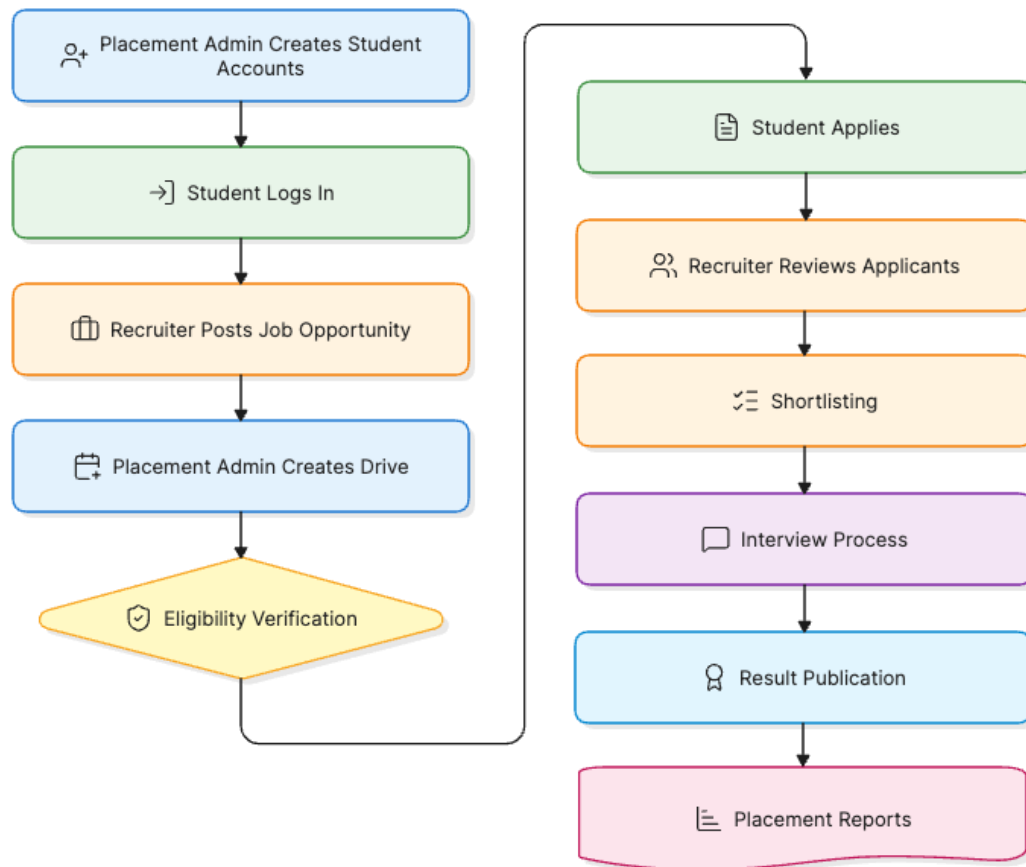
The system automates the complete placement lifecycle from student onboarding to final placement reporting.

System Architecture Design



4.2 Project Functions

Placement Workflow Activity Diagram



The system shall provide the following functionalities:

Student Account Management

- Create student accounts
- Update student records
- Activate/deactivate student accounts
- Reset student credentials

Student Profile Management

- Maintain personal information
- Maintain academic records
- Upload resumes

Recruiter Management

- Recruiter registration
- Company profile management
- Recruiter account management

Placement Drive Management

- Create drives
- Schedule recruitment activities
- Publish placement opportunities

Eligibility Management

- Define eligibility criteria
- Verify student eligibility automatically

Application Management

- Apply for placement drives
- Track application status

Notification Management

- Drive announcements
- Interview notifications
- Result notifications

Reporting & Analytics

- Placement statistics
- Student reports
- Company reports
- Department-wise reports
- Year-wise reports

4.3 User Classes and Characteristics

Super Admin

Responsibilities:

- Manage Placement Admin accounts
- Configure system settings
- Manage permissions
- Monitor system activities
- Generate administrative reports

Placement Admin

Responsibilities:

- Create and manage student accounts
- Manage recruiter accounts
- Create placement drives
- Define eligibility criteria
- Publish announcements
- Monitor applications
- Generate reports

- Publish placement results

Recruiter / Company HR

Responsibilities:

- Maintain company profile
- Post job opportunities
- Define eligibility criteria
- View applicants
- Shortlist candidates
- Update recruitment status
- Publish results

Student

Responsibilities:

- Login to the portal
- Maintain profile
- Upload resume
- View placement opportunities
- Apply for eligible drives
- Track application status
- View notifications and results

4.4 Operating Environment

4.4.1 Server Environment

Component	Technology
Operating System	Windows 10 / Windows 11
Programming Language	Java
Backend Framework	Spring Boot 3
Database	MySQL
Authentication	JWT
Build Tool	Maven
Version Control	Git

4.4.2 Client Environment

Supported Browsers:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Supported Operating Systems:

- Windows
- Linux
- MacOS

Mobile Support:

- Responsive Web Interface
- Android Mobile Browsers

4.5 Design and Implementation Constraints

- Internet connectivity is required.
- Student accounts can only be created by Placement Admins.
- Role-based access control must be enforced.
- Passwords must be encrypted.
- Browser compatibility must be maintained.
- Secure authentication mechanisms must be implemented.

4.6 User Documentation

The application will include documentation to help users understand and use the system effectively. This documentation will contain:

- A user guide explaining the main features of the system and how to use them.
- Instructions for setting up and accessing the application for the first time.
- A FAQ section addressing common questions and issues.
- Details on how users can contact support if additional assistance is required.

This documentation will be made available through the application and/or as a downloadable document.

4.7 Assumptions and Dependencies

Assumptions

- Students belong to verified institutions.
- Recruiters provide authentic company information.
- Users have internet access.

Dependencies

- React Frontend Application
- Spring Boot Backend Services
- MySQL Database Server
- Email Notification Service

5.0 External Interface Requirements

5.1 User Interfaces

Authentication Module

- Login Page
- Forgot Password Page
- Change Password Page

Student Module

- Dashboard
- Profile Management
- Resume Upload
- Placement Drives
- Application Status

Placement Admin Module

- Dashboard
- Student Management
- Recruiter Management
- Placement Drive Management
- Reports & Analytics

Recruiter Module

- Recruiter Dashboard
- Job Posting
- Applicant Management
- Candidate Shortlisting
- Result Publishing

Super Admin Module

- User Management
- Access Control
- System Monitoring
- Reports

5.2 System Interfaces

The Proposed system configuration is as follows:

S. No.	Item	Specification
01	Processor	Quad core processor (64 bit)
02	No. of processors	1
03	Memory / RAM	8 GB (Minimum)
04	Storage	1 TB
05	Network	Gigabit Ethernet
06	OS	Windows 10 / Windows 11

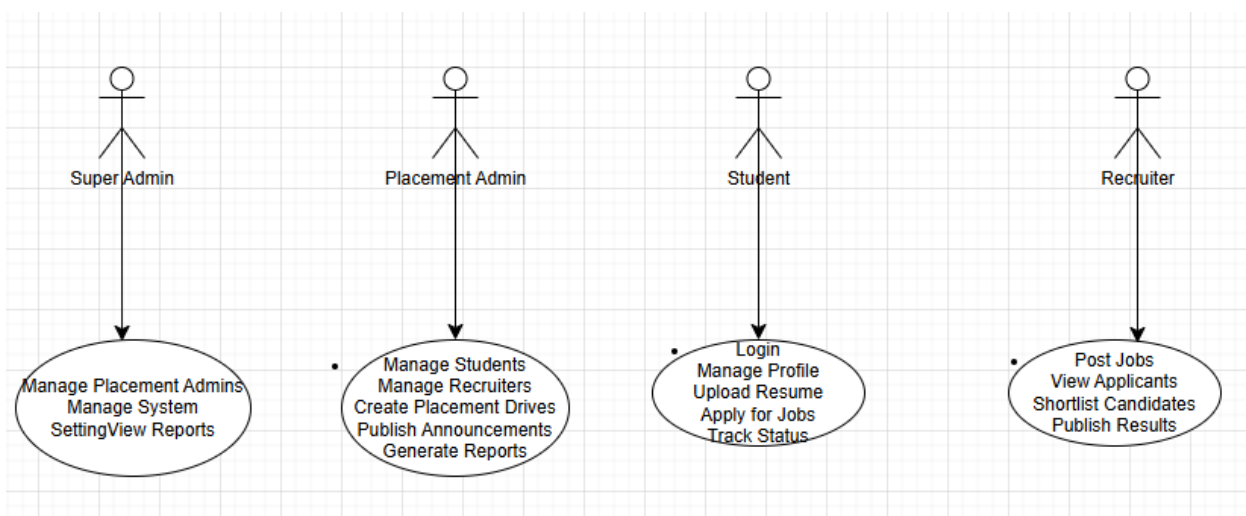
5.3 Communication Interfaces

The system shall support:

- HTTP/HTTPS Communication
- RESTful APIs
- JWT Authentication
- Email Notifications

6.0 System Features (Functional Requirements)

6.1 Use Case Diagram



Feature 1: Authentication and Authorization

Actors

- Super Admin
- Placement Admin
- Recruiter
- Student

FR-1.1: User Login

- **ID:** FR-1.1
- **Title:** User Login
- **Description:** The system shall allow Super Admin, Placement Admin, Recruiter, and Student users to log in using valid credentials. Upon successful authentication, users shall be redirected to their respective dashboards based on their assigned roles.
- **Rationale:** Required to provide secure access to the system and ensure users can access authorized functionalities.
- **Inputs:** Username/Email, Password
- **Outputs:** JWT Token, Dashboard Access, Error Message (if authentication fails)
- **Preconditions:** User account exists and is active.
- **Postconditions:** User session is established, JWT token is generated, and login activity is recorded.

Acceptance Criteria:

1. Valid credentials → User authenticated and redirected to role-specific dashboard.
 2. Invalid credentials → Display "Invalid username or password".
 3. Inactive account → Access denied with appropriate message.
 4. JWT token generated successfully after login.
- **Priority:** High
 - **Dependencies:** Users Database, JWT Authentication Service

FR-1.2: User Logout

- **ID:** FR-1.2
- **Title:** User Logout
- **Description:** The system shall allow authenticated users to securely log out from the application and invalidate their active session/token.
- **Rationale:** Required to prevent unauthorized access after a user finish using the system.
- **Inputs:** Logout Request
- **Outputs:** Logout Confirmation, Redirect to Login Page
- **Preconditions:** User is authenticated and logged into the system.
- **Postconditions:** User session is terminated and JWT token is invalidated.

Acceptance Criteria:

1. Authenticated user clicks logout → Session terminated successfully.
2. User redirected to login page.
3. Accessing protected resources after logout requires re-authentication.

- **Priority:** High
- **Dependencies:** User Login, JWT Authentication Service

FR-1.3: Password Reset

- **ID:** FR-1.3
- **Title:** Password Reset
- **Description:** The system shall allow registered users to reset their password through a secure password recovery mechanism.
- **Rationale:** Required to enable users to regain access to their accounts when they forget their passwords.
- **Inputs:** Registered Email Address, New Password, Reset Token/OTP
- **Outputs:** Password Reset Confirmation or Error Message
- **Preconditions:** User account exists, and password reset request has been initiated.
- **Postconditions:** User password is updated successfully in the database.

Acceptance Criteria:

1. Valid reset token/OTP → Password updated successfully.
2. Invalid or expired token/OTP → Display appropriate error message.
3. User can log in using the new password after reset.

- **Priority:** High
- **Dependencies:** User Database, Email/OTP Service

FR-1.4: Role-Based Access Control (RBAC)

- **ID:** FR-1.4
- **Title:** Role-Based Access Control
- **Description:** The system shall restrict access to functionalities and resources based on the user's assigned role (Super Admin, Placement Admin, Recruiter, or Student).
- **Rationale:** Required to ensure users can only access features relevant to their responsibilities and permissions.
- **Inputs:** Authenticated User Role
- **Outputs:** Authorized Access or Access Denied Message
- **Preconditions:** User is authenticated and assigned a valid role.
- **Postconditions:** User gains access only to permitted modules and operations.

Acceptance Criteria:

1. Super Admin can access all system functionalities.
2. Placement Admin can access placement management functions.
3. Recruiter can access recruitment-related functions only.
4. Student can access student-related functions only.
5. Unauthorized access attempts result in an "Access Denied" message.

- **Priority:** High
- **Dependencies:** User Login, User Roles Database

FR-1.5: JWT Authentication

- **ID:** FR-1.5
- **Title:** JWT-Based Authentication
- **Description:** The system shall use JSON Web Tokens (JWT) to authenticate and authorize users for accessing protected resources.
- **Rationale:** Required to provide secure, stateless authentication for web application requests.
- **Inputs:** Valid User Credentials, JWT Token
- **Outputs:** Authentication Token, Access Authorization, Error Message
- **Preconditions:** User has successfully logged in.
- **Postconditions:** JWT token is generated, validated for subsequent requests, and used to authorize access to protected endpoints.

Acceptance Criteria:

1. Successful login generates a valid JWT token.
 2. Requests with valid JWT tokens are authorized.
 3. Requests with expired, invalid, or missing tokens are rejected.
 4. Protected APIs are inaccessible without a valid JWT token.
- **Priority:** High
 - **Dependencies:** User Login, Security Framework, User Database

Feature 2: Student Account Management

Actor

Placement Admin

FR-2.1: Create Student Account

- **ID:** FR-2.1
- **Title:** Create Student Account

- **Description:** The system shall allow authorized Placement Admins to create student accounts by entering the required student information. Students shall not be allowed to self-register.
- **Rationale:** Required to ensure that only verified students are granted access to the placement portal.
- **Inputs:** Student Name, Roll Number, Email Address, Department, Batch, Contact Details
- **Outputs:** Student Account Creation Confirmation, Login Credentials
- **Preconditions:** Placement Admin is authenticated and authorized. Student account does not already exist.
- **Postconditions:** Student account is created and stored in the database.

Acceptance Criteria:

1. Placement Admin can successfully create a new student account.
2. Duplicate student records are rejected.
3. System generates credentials for the student account.
4. Created account is available for login.

- **Priority:** High
- **Dependencies:** Authentication and Authorization Module, Student Database

FR-2.2: Update Student Details

- **ID:** FR-2.2
- **Title:** Update Student Details
- **Description:** The system shall allow Placement Admins to update student information including academic, personal, and contact details.
- **Rationale:** Required to maintain accurate and up-to-date student records.
- **Inputs:** Student ID, Updated Student Information
- **Outputs:** Update Confirmation Message
- **Preconditions:** Placement Admin is authenticated. Student account exists.
- **Postconditions:** Student information is updated in the database.

Acceptance Criteria:

1. Placement Admin can modify existing student details.
2. Updated information is reflected immediately.
3. Invalid data entries are rejected with appropriate error messages.

- **Priority:** High
- **Dependencies:** Student Database

FR-2.3: Activate Student Account

- **ID:** FR-2.3
- **Title:** Activate Student
- **Description:** The system shall allow Placement Admins to activate student accounts, enabling students to access portal functionalities.
- **Rationale:** Required to control student access to the placement portal.
- **Inputs:** Student ID
- **Outputs:** Account Activation Confirmation
- **Preconditions:** Student account exists and is currently inactive.
- **Postconditions:** Student account status is changed to Active.

Acceptance Criteria:

1. Inactive student account can be activated successfully.
 2. Activated student can log in to the system.
 3. Activation status is reflected in student records.
- **Priority:** Medium
 - **Dependencies:** Student Account Management Module

FR-2.4: Deactivate Student Account

- **ID:** FR-2.4
- **Title:** Deactivate Student Account
- **Description:** The system shall allow Placement Admins to deactivate student accounts, preventing access to the portal.
- **Rationale:** Required for handling graduated, inactive, or unauthorized students.
- **Inputs:** Student ID
- **Outputs:** Account Deactivation Confirmation
- **Preconditions:** Student account exists and is active.
- **Postconditions:** Student account status is changed to Inactive and login access is disabled.

Acceptance Criteria:

1. Active student account can be deactivated successfully.
 2. Deactivated students cannot log in.
 3. Account status is updated in the database.
- **Priority:** Medium
 - **Dependencies:** Student Account Management Module

FR-2.5: Reset Student Credentials

- **ID:** FR-2.5
- **Title:** Reset Student Credentials
- **Description:** The system shall allow Placement Admins to reset a student's login credentials when required.
- **Rationale:** Required to assist students who have lost or forgotten their login credentials.
- **Inputs:** Student ID
- **Outputs:** New Credentials or Password Reset Confirmation
- **Preconditions:** Placement Admin is authenticated. Student account exists.
- **Postconditions:** Student credentials are reset, and previous credentials become invalid.

Acceptance Criteria:

1. Placement Admin can initiate credential reset.
 2. New credentials are generated, or password reset is completed successfully.
 3. Previous credentials can no longer be used for login.
- **Priority:** High
 - **Dependencies:** Authentication Module, Student Database

Feature 3: Student Profile Management

Actor

Student

FR-3.1: Update Personal Information

- **ID:** FR-3.1
- **Title:** Update Personal Information
- **Description:** The system shall allow students to update their personal information, including contact details and other relevant profile information.
- **Rationale:** Required to ensure that student records remain accurate and up to date for placement activities.
- **Inputs:** Student ID, Personal Information (Name, Contact Number, Email, Address, etc.)
- **Outputs:** Profile Update Confirmation Message
- **Preconditions:** Student is authenticated and logged into the system.
- **Postconditions:** Updated personal information is stored in the database.

Acceptance Criteria:

1. Student can modify editable personal information fields.
2. Changes are saved successfully and reflected in the profile.
3. Invalid input values are rejected with appropriate error messages.

- **Priority:** High
- **Dependencies:** Authentication Module, Student Database

FR-3.2: Update Academic Information

ID: FR-3.2

Title: Update Academic Information

Description: The system shall allow students to update their academic details, including CGPA, skills, certifications, and educational information.

Rationale: Required to maintain accurate academic records for eligibility verification and recruiter access.

Inputs: Student ID, Academic Details

Outputs: Academic Information Update Confirmation

Preconditions: Student is authenticated and logged into the system.

Postconditions: Academic information is updated in the database.

Acceptance Criteria:

- Student can update academic information successfully.
- Updated information is immediately visible in the profile.
- Invalid or incomplete academic data is rejected.

Priority: High

Dependencies: Student Database

FR-3.3: Upload Resume

- **ID:** FR-3.3
- **Title:** Upload Resume
- **Description:** The system shall allow students to upload and maintain their latest resume for placement applications.
- **Rationale:** Required to provide recruiters with access to current student resumes during recruitment processes.
- **Inputs:** Resume File (PDF/DOC/DOCX)
- **Outputs:** Resume Upload Confirmation
- **Preconditions:** Student is authenticated and logged into the system.

- **Postconditions:** Resume file is stored and linked to the student profile.

Acceptance Criteria:

1. Student can upload a valid resume file.
 2. Uploaded resume is stored successfully.
 3. Existing resume can be replaced with a newer version.
 4. Unsupported file formats are rejected.
- **Priority:** High
 - **Dependencies:** File Storage Service, Student Database

FR-3.4: View Profile

- **ID:** FR-3.4
- **Title:** View Profile
- **Description:** The system shall allow students to view their complete profile, including personal information, academic details, and uploaded resume.
- **Rationale:** Required to enable students to verify and review the information available to recruiters and placement administrators.
- **Inputs:** Student Profile Request
- **Outputs:** Student Profile Details
- **Preconditions:** Student is authenticated and logged into the system.
- **Postconditions:** Student profile information is displayed successfully.

Acceptance Criteria:

1. Student can view all profile details.
 2. Profile information displayed matches the stored data.
 3. Uploaded resume information is accessible from the profile page.
- **Priority:** Medium
 - **Dependencies:** Authentication Module, Student Database, Resume Management Module

Feature 4: Recruiter Management

Actors

- Placement Admin

FR-4.1: Add Recruiter

- **ID:** FR-4.1
- **Title:** Add Recruiter

- **Description:** The system shall allow Placement Admins to create recruiter accounts by entering company and recruiter details.
- **Rationale:** Required to enable authorized recruiters to participate in campus recruitment activities through the portal.
- **Inputs:** Company Name, Recruiter Name, Email Address, Contact Number, Company Details
- **Outputs:** Recruiter Account Creation Confirmation, Login Credentials
- **Preconditions:** Placement Admin is authenticated and authorized. Recruiter account does not already exist.
- **Postconditions:** Recruiter account is created and stored in the database.

Acceptance Criteria:

1. Placement Admin can create a recruiter account successfully.
2. Duplicate recruiter accounts are rejected.
3. Login credentials are generated for the recruiter.
4. Newly created recruiter can access the system after activation.

- **Priority:** High
- **Dependencies:** Authentication Module, Recruiter Database

FR-4.2: Update Recruiter Information

- **ID:** FR-4.2
- **Title:** Update Recruiter Information
- **Description:** The system shall allow Placement Admins to update recruiter and company-related information.
- **Rationale:** Required to maintain accurate recruiter records and company details.
- **Inputs:** Recruiter ID, Updated Recruiter Information, Updated Company Details
- **Outputs:** Update Confirmation Message
- **Preconditions:** Placement Admin is authenticated. Recruiter account exists.
- **Postconditions:** Recruiter information is updated in the database.

Acceptance Criteria:

1. Placement Admin can modify recruiter information.
2. Updated information is reflected immediately.
3. Invalid data entries are rejected with appropriate error messages.

- **Priority:** Medium
- **Dependencies:** Recruiter Database

FR-4.3: Manage Recruiter Accounts

- **ID:** FR-4.3
- **Title:** Manage Recruiter Accounts
- **Description:** The system shall allow Placement Admins to manage recruiter accounts, including activating, deactivating, and resetting recruiter credentials.
- **Rationale:** Required to control recruiter access and ensure account security.
- **Inputs:** Recruiter ID, Management Action (Activate, Deactivate, Reset Credentials)
- **Outputs:** Account Management Confirmation Message
- **Preconditions:** Placement Admin is authenticated. Recruiter account exists.
- **Postconditions:** Recruiter account status or credentials are updated accordingly.

Acceptance Criteria:

1. Placement Admin can activate inactive recruiter accounts.
 2. Placement Admin can deactivate active recruiter accounts.
 3. Placement Admin can reset recruiter credentials.
 4. Deactivated recruiters cannot access the system.
 5. Changes are reflected immediately in the recruiter account status.
- **Priority:** High
 - **Dependencies:** Authentication Module, Recruiter Database, Account Management Module

Feature 5: Placement Drive Management

Actor

- Placement Admin

FR-5.1: Create Placement Drive

- **ID:** FR-5.1
- **Title:** Create Placement Drive
- **Description:** The system shall allow Placement Admins to create placement drives by entering company details, job information, recruitment schedules, and other relevant placement data.
- **Rationale:** Required to organize and manage campus recruitment activities efficiently.
- **Inputs:** Company Name, Job Title, Job Description, Recruitment Dates, Drive Details
- **Outputs:** Placement Drive Creation Confirmation
- **Preconditions:** Placement Admin is authenticated and authorized.
- **Postconditions:** Placement drive is created and stored in the system.

Acceptance Criteria:

1. Placement Admin can create a new placement drive successfully.
2. Mandatory placement drive details are validated before creation.
3. Created placement drive is available for further configuration.
4. Duplicate placement drive entries are prevented where applicable.

- **Priority:** High
- **Dependencies:** Authentication Module, Recruiter Management

FR-5.2: Define Eligibility Criteria

- **ID:** FR-5.2
- **Title:** Define Eligibility Criteria
- **Description:** The system shall allow Placement Admins to define eligibility criteria for placement drives, including CGPA, department, graduation year, and other recruitment requirements.
- **Rationale:** Required to ensure only qualified students are allowed to apply for opportunities.
- **Inputs:** Placement Drive ID, Eligibility Criteria
- **Outputs:** Eligibility Criteria Configuration Confirmation
- **Preconditions:** Placement drive exists in the system.
- **Postconditions:** Eligibility criteria are associated with the placement drive.

Acceptance Criteria:

1. Placement Admin can define one or more eligibility requirements.
2. Eligibility criteria are stored successfully.
3. Criteria can be modified before drive publication.
4. Defined criteria are used during eligibility verification.

- **Priority:** High
- **Dependencies:** Placement Drive Management, Eligibility Verification Module

FR-5.3: Schedule Recruitment Activities

- **ID:** FR-5.3
- **Title:** Schedule Recruitment Activities
- **Description:** The system shall allow Placement Admins to schedule recruitment activities such as aptitude tests, interviews, group discussions, and other selection rounds.
- **Rationale:** Required to coordinate recruitment events and communicate schedules to participants.
- **Inputs:** Placement Drive ID, Activity Type, Date, Time, Venue/Mode
- **Outputs:** Activity Schedule Confirmation
- **Preconditions:** Placement drive exists in the system.

- **Postconditions:** Recruitment activities are scheduled and stored in the system.

Acceptance Criteria:

1. Placement Admin can schedule recruitment activities successfully.
 2. Scheduled activities include date, time, and location details.
 3. Activity schedules can be modified if required.
 4. Scheduled activities are visible to eligible participants.
- **Priority:** High
 - **Dependencies:** Placement Drive Management, Notification Management

FR-5.4: Publish Placement Opportunities

- **ID:** FR-5.4
- **Title:** Publish Placement Opportunities
- **Description:** The system shall allow Placement Admins to publish placement drives and job opportunities for eligible students.
- **Rationale:** Required to make placement opportunities available to students and initiate the application process.
- **Inputs:** Placement Drive ID, Publication Status
- **Outputs:** Published Placement Opportunity
- **Preconditions:** Placement drive has been created and configured with eligibility criteria.
- **Postconditions:** Placement opportunity becomes visible to eligible students.

Acceptance Criteria:

1. Placement Admin can publish placement opportunities successfully.
 2. Only published opportunities are visible to students.
 3. Eligibility criteria are enforced for opportunity visibility and applications.
 4. Students receive notifications regarding newly published opportunities.
- **Priority:** High
 - **Dependencies:** Placement Drive Management, Eligibility Verification Module, Notification Management Module

Feature 6: Eligibility Verification

Actors

- Placement Admin
- Recruiter

FR-6.1: Verify Student Eligibility Based on CGPA

- **ID:** FR-6.1
- **Title:** Verify Student Eligibility Based on CGPA
- **Description:** The system shall automatically verify whether a student's CGPA meets the minimum CGPA requirement specified for a job posting before allowing the student to apply.
- **Rationale:** Required to ensure that only qualified students can apply for job opportunities.
- **Inputs:** Student CGPA, Job CGPA Requirement
- **Outputs:** Eligibility Status (Eligible/Not Eligible)
- **Preconditions:** Student profile and job posting exist in the system.
- **Postconditions:** Eligibility result is generated, and application access is granted or denied accordingly.

Acceptance Criteria:

1. Students meeting the CGPA requirement are marked as eligible.
 2. Students below the CGPA requirement are marked as not eligible.
 3. Ineligible students cannot submit applications.
 4. Eligibility status is displayed before application submission.
- **Priority:** High
 - **Dependencies:** Student Profile Management, Job Posting Management

FR-6.2: Verify Student Eligibility Based on Department

- **ID:** FR-6.2
- **Title:** Verify Student Eligibility Based on Department
- **Description:** The system shall automatically verify whether a student's department matches the departments specified in a job posting.
- **Rationale:** Required to ensure applications are restricted to eligible academic disciplines.
- **Inputs:** Student Department, Eligible Departments for Job Posting
- **Outputs:** Eligibility Status (Eligible/Not Eligible)
- **Preconditions:** Student profile and job posting exist.
- **Postconditions:** Department eligibility is validated before application submission.

Acceptance Criteria:

1. Students from eligible departments can proceed with applications.
2. Students from non-eligible departments are restricted from applying.
3. Eligibility status is displayed clearly to the student.

- **Priority:** High
- **Dependencies:** Student Profile Management, Job Posting Management

FR-6.3: Verify Student Eligibility Based on Graduation Year

- **ID:** FR-6.3
- **Title:** Verify Student Eligibility Based on Graduation Year
- **Description:** The system shall automatically verify whether a student's graduation year matches the graduation year requirements defined for a job posting.
- **Rationale:** Required to ensure only students from eligible graduating batches can apply.
- **Inputs:** Student Graduation Year, Eligible Graduation Year(s)
- **Outputs:** Eligibility Status (Eligible/Not Eligible)
- **Preconditions:** Student profile and job posting exist.
- **Postconditions:** Graduation year eligibility is validated before application submission.

Acceptance Criteria:

1. Students belonging to eligible graduation years can apply.
2. Students from non-eligible graduation years are restricted from applying.
3. Eligibility status is displayed before application submission.

- **Priority:** High
- **Dependencies:** Student Profile Management, Job Posting Management

FR-6.4: Verify Student Eligibility Based on Company-Specific Requirements

- **ID:** FR-6.4
- **Title:** Verify Student Eligibility Based on Company-Specific Requirements
- **Description:** The system shall automatically evaluate additional eligibility requirements defined by recruiters, such as skills, certifications, academic criteria, or other custom requirements.
- **Rationale:** Required to support diverse recruitment criteria established by different companies.
- **Inputs:** Student Profile Information, Company-Specific Eligibility Requirements
- **Outputs:** Eligibility Status (Eligible/Not Eligible)
- **Preconditions:** Recruiter has defined company-specific eligibility criteria. Student profile is available.
- **Postconditions:** Eligibility validation result is generated, and application access is granted or denied accordingly.

Acceptance Criteria:

1. Students meeting all company-specific requirements are marked as eligible.
2. Students failing any mandatory requirement are marked as not eligible.
3. Eligibility results are generated automatically during application submission.

4. Ineligible students cannot proceed with the application process.

- **Priority:** High
- **Dependencies:** Student Profile Management, Recruiter Management, Job Posting Management

FR-6.5: Overall Eligibility Validation

- **ID:** FR-6.5
- **Title:** Overall Eligibility Validation
- **Description:** The system shall evaluate all applicable eligibility criteria, including CGPA, Department, Graduation Year, and Company-Specific Requirements, before allowing a student to apply for a job.
- **Rationale:** Required to ensure a consistent and automated eligibility verification process.
- **Inputs:** Student Profile Data, Job Eligibility Criteria
- **Outputs:** Final Eligibility Decision (Eligible/Not Eligible)
- **Preconditions:** Student profile and job posting exist.
- **Postconditions:** Student is either allowed or restricted from submitting a job application.

Acceptance Criteria:

1. Student is marked eligible only if all mandatory criteria are satisfied.
 2. Student is prevented from applying if any mandatory criterion fails.
 3. Eligibility validation occurs automatically before application submission.
 4. The reason for ineligibility is displayed to the student.
- **Priority:** High
 - **Dependencies:** FR-6.1, FR-6.2, FR-6.3, FR-6.4, Student Profile Management, Job Posting Management

Feature 7: Job Application Management

Actor

- Student

FR-7.1: View Eligible Drives

- **ID:** FR-7.1
- **Title:** View Eligible Drives
- **Description:** The system shall allow students to view all placement drives and job opportunities for which they meet the eligibility criteria.
- **Rationale:** Required to help students identify relevant placement opportunities without manually checking eligibility requirements.

- **Inputs:** Student Profile Information
- **Outputs:** List of Eligible Placement Drives and Job Opportunities
- **Preconditions:** Student is authenticated, and eligibility criteria have been evaluated.
- **Postconditions:** Eligible opportunities are displayed to the student.

Acceptance Criteria:

1. Student can view only opportunities for which they are eligible.
 2. Each opportunity displays job details and eligibility criteria.
 3. Ineligible opportunities are hidden or clearly marked as unavailable.
 4. Opportunity list is updated whenever eligibility criteria change.
- **Priority:** High
 - **Dependencies:** Eligibility Verification Module, Job Posting Management

FR-7.2: Apply for Opportunities

- **ID:** FR-7.2
- **Title:** Apply for Opportunities
- **Description:** The system shall allow eligible students to submit applications for placement drives and job opportunities.
- **Rationale:** Required to enable students to participate in recruitment processes through the portal.
- **Inputs:** Job Opportunity Selection, Student Resume, Application Submission Request
- **Outputs:** Application Confirmation Message
- **Preconditions:** Student is authenticated, profile is complete, and eligibility requirements are satisfied.
- **Postconditions:** Application record is created, and its status is set to "Applied".

Acceptance Criteria:

1. Eligible students can successfully submit applications.
 2. Ineligible students cannot apply.
 3. Duplicate applications for the same opportunity are prevented.
 4. Application status is automatically set to "Applied" after submission.
- **Priority:** High
 - **Dependencies:** Student Profile Management, Eligibility Verification, Resume Management

FR-7.3: Track Application Status

- **ID:** FR-7.3
- **Title:** Track Application Status
- **Description:** The system shall allow students to view and track the status of all submitted job applications.
- **Rationale:** Required to provide transparency throughout the recruitment process.
- **Inputs:** Student Application Request
- **Outputs:** Current Application Status and Related Updates
- **Preconditions:** Student is authenticated and has submitted at least one application.
- **Postconditions:** Application status information is displayed to the student.

Acceptance Criteria:

1. Student can view all submitted applications.
 2. Status of each application is displayed.
 3. Status updates are reflected immediately after changes are made.
 4. Students can view relevant recruitment progress information.
- **Priority:** High
 - **Dependencies:** Job Application Management, Recruiter Management

FR-7.4: Application Status Management – Applied

- **ID:** FR-7.4
- **Title:** Application Status – Applied
- **Description:** The system shall assign the status "Applied" when a student successfully applies for a job opportunity.
- **Rationale:** Required to indicate successful application submission.
- **Inputs:** Application Submission Request
- **Outputs:** Application Status = Applied
- **Preconditions:** Student application is successfully submitted.
- **Postconditions:** Application record is stored with status "Applied".

Acceptance Criteria:

1. Status is automatically assigned after successful submission.
 2. Student can view the "Applied" status in the application tracking section.
- **Priority:** High
 - **Dependencies:** FR-7.2 Apply for Opportunities

FR-7.5: Application Status Management – Shortlisted

- **ID:** FR-7.5
- **Title:** Application Status – Shortlisted
- **Description:** The system shall update an application's status to "Shortlisted" when the recruiter selects the student for the next stage of the recruitment process.
- **Rationale:** Required to notify students that they have progressed in the selection process.
- **Inputs:** Recruiter Shortlisting Action
- **Outputs:** Application Status = Shortlisted
- **Preconditions:** Application status is "Applied".
- **Postconditions:** Student application status is updated to "Shortlisted".

Acceptance Criteria:

1. Recruiter can shortlist eligible applicants.
 2. Student can view the updated status.
 3. Status history is maintained.
- **Priority:** High
 - **Dependencies:** Recruiter Management, Application Management

FR-7.6: Application Status Management – Interview Scheduled

- **ID:** FR-7.6
- **Title:** Application Status – Interview Scheduled
- **Description:** The system shall update an application's status to "Interview Scheduled" when an interview is arranged for the shortlisted student.
- **Rationale:** Required to inform students about upcoming interview activities.
- **Inputs:** Interview Schedule Details
- **Outputs:** Application Status = Interview Scheduled
- **Preconditions:** Application status is "Shortlisted".
- **Postconditions:** Interview details and status are updated in the system.

Acceptance Criteria:

1. Interview schedule can be assigned to shortlisted students.
 2. Student can view interview details and updated status.
 3. Schedule information is stored successfully.
- **Priority:** High
 - **Dependencies:** Interview Management, Recruiter Management

FR-7.7: Application Status Management – Selected

- **ID:** FR-7.7
- **Title:** Application Status – Selected
- **Description:** The system shall update an application's status to "Selected" when the student successfully clears all recruitment stages and receives an offer.
- **Rationale:** Required to record successful recruitment outcomes.
- **Inputs:** Recruiter Selection Decision
- **Outputs:** Application Status = Selected
- **Preconditions:** Recruitment process is completed successfully.
- **Postconditions:** Student application status is updated to "Selected".

Acceptance Criteria:

1. Recruiter can mark a candidate as selected.
 2. Student can view the selection status.
 3. Selection records are stored for reporting purposes.
- **Priority:** High
 - **Dependencies:** Recruiter Management, Application Management

FR-7.8: Application Status Management – Rejected

- **ID:** FR-7.8
- **Title:** Application Status – Rejected
- **Description:** The system shall update an application's status to "Rejected" when the student is not selected during any stage of the recruitment process.
- **Rationale:** Required to provide final application outcomes and maintain accurate recruitment records.
- **Inputs:** Recruiter Rejection Decision
- **Outputs:** Application Status = Rejected
- **Preconditions:** Student application exists in the recruitment process.
- **Postconditions:** Student application status is updated to "Rejected".

Acceptance Criteria:

1. Recruiter can mark an application as rejected.
 2. Student can view the rejection status.
 3. Rejection records are maintained for future reference.
- **Priority:** High
 - **Dependencies:** Recruiter Management, Application Management

Feature 8: Candidate Shortlisting

Actor

- Recruiter

Functions

FR-8.1: View Applicants

- **ID:** FR-8.1
- **Title:** View Applicants
- **Description:** The system shall allow recruiters to view all students who have applied for a specific job opportunity or placement drive.
- **Rationale:** Required to enable recruiters to review candidate applications and begin the selection process.
- **Inputs:** Job Posting Selection
- **Outputs:** List of Applicants with Profile Details and Resumes
- **Preconditions:** Recruiter is authenticated and at least one application exists for the selected job posting.
- **Postconditions:** Applicant information is displayed to the recruiter.

Acceptance Criteria:

1. Recruiter can view all applicants for a selected job posting.
 2. Applicant profiles and resumes are accessible.
 3. Applicant information is displayed accurately.
 4. Only applications related to the recruiter's job postings are visible.
- **Priority:** High
 - **Dependencies:** Job Application Management, Student Profile Management

FR-8.2: Filter Candidates

- **ID:** FR-8.2
- **Title:** Filter Candidates
- **Description:** The system shall allow recruiters to filter applicants based on criteria such as CGPA, department, graduation year, skills, certifications, and other profile attributes.
- **Rationale:** Required to simplify the candidate screening and selection process.
- **Inputs:** Filter Criteria
- **Outputs:** Filtered List of Candidates
- **Preconditions:** Applicant records exist for the selected job posting.
- **Postconditions:** Matching candidates are displayed according to the selected criteria.

Acceptance Criteria:

1. Recruiter can apply one or more filtering criteria.
 2. Candidate list updates based on selected filters.
 3. Recruiter can clear filters and view all applicants.
 4. Filter results are generated accurately.
- **Priority:** High
 - **Dependencies:** Student Profile Management, Job Application Management

FR-8.3: Shortlist Students

- **ID:** FR-8.3
- **Title:** Shortlist Students
- **Description:** The system shall allow recruiters to shortlist applicants who meet the required selection criteria for further recruitment stages.
- **Rationale:** Required to identify candidates eligible for interviews and subsequent recruitment activities.
- **Inputs:** Candidate Selection, Shortlisting Decision
- **Outputs:** Application Status Updated to "Shortlisted"
- **Preconditions:** Candidate has applied for the job opportunity.
- **Postconditions:** Candidate application status is updated to "Shortlisted".

Acceptance Criteria:

1. Recruiter can shortlist one or more applicants.
 2. Application status changes to "Shortlisted".
 3. Students can view the updated status.
 4. Shortlisted candidate records are maintained.
- **Priority:** High
 - **Dependencies:** Job Application Management

FR-8.4: Update Interview Status

- **ID:** FR-8.4
- **Title:** Update Interview Status
- **Description:** The system shall allow recruiters to schedule interviews and update the interview status of shortlisted candidates.
- **Rationale:** Required to track candidate progress through the interview process.
- **Inputs:** Candidate Selection, Interview Details, Status Update
- **Outputs:** Updated Interview Status and Schedule Information
- **Preconditions:** Candidate application status is "Shortlisted".

- **Postconditions:** Interview details and status are stored and made available to students.

Acceptance Criteria:

1. Recruiter can schedule interviews for shortlisted candidates.
 2. Recruiter can update interview-related statuses.
 3. Students can view interview schedules and status updates.
 4. Interview information is stored successfully.
- **Priority:** High
 - **Dependencies:** Candidate Shortlisting, Job Application Management

FR-8.5: Publish Results

- **ID:** FR-8.5
- **Title:** Publish Recruitment Results
- **Description:** The system shall allow recruiters to publish final recruitment outcomes by marking candidates as Selected or Rejected.
- **Rationale:** Required to communicate recruitment decisions and maintain placement records.
- **Inputs:** Candidate Selection Decision (Selected/Rejected)
- **Outputs:** Updated Application Status, Recruitment Results
- **Preconditions:** Recruitment process for the candidate has been completed.
- **Postconditions:** Candidate application status is updated and results are visible to students and placement administrators.

Acceptance Criteria:

1. Recruiter can mark candidates as Selected or Rejected.
 2. Application status is updated immediately.
 3. Students can view final recruitment results.
 4. Published results are stored for reporting and analytics purposes.
- **Priority:** High
 - **Dependencies:** Job Application Management, Interview Management, Recruiter Management

Feature 9: Notification Management

Actors

- Placement Admin
- Recruiter

FR-9.1: Placement Announcements

- **ID:** FR-9.1
- **Title:** Placement Announcements
- **Description:** The system shall allow Placement Admins and Recruiters to publish placement-related announcements such as new job opportunities, placement drives, deadlines, and important updates.
- **Rationale:** Required to keep students informed about placement activities and opportunities.
- **Inputs:** Announcement Title, Announcement Content, Target Audience, Publication Date
- **Outputs:** Published Announcement Notification
- **Preconditions:** User is authenticated as Placement Admin or Recruiter.
- **Postconditions:** Announcement is published and visible to intended recipients.

Acceptance Criteria:

1. Placement Admin and Recruiter can create announcements.
 2. Published announcements are visible to eligible students.
 3. Announcement details are displayed accurately.
 4. Announcements remain accessible until removed or expired.
- **Priority:** High
 - **Dependencies:** Authentication Module, Job Posting Management

FR-9.2: Interview Notifications

- **ID:** FR-9.2
- **Title:** Interview Notifications
- **Description:** The system shall automatically notify students when interview schedules are created or updated by recruiters.
- **Rationale:** Required to ensure students receive timely information regarding interview schedules and arrangements.
- **Inputs:** Interview Schedule Details, Candidate Information
- **Outputs:** Interview Notification Message
- **Preconditions:** Interview has been scheduled or modified for a shortlisted candidate.
- **Postconditions:** Notification is delivered to the concerned student.

Acceptance Criteria:

1. Notification is generated when an interview is scheduled.
 2. Notification includes interview date, time, and relevant details.
 3. Students can view interview notifications within the system.
 4. Notification is sent only to affected candidates.
- **Priority:** High
 - **Dependencies:** Candidate Shortlisting, Interview Management

FR-9.3: Result Notifications

- **ID:** FR-9.3
- **Title:** Result Notifications
- **Description:** The system shall automatically notify students when recruitment results are published, or application statuses are updated.
- **Rationale:** Required to provide timely communication of recruitment outcomes.
- **Inputs:** Application Status Update, Recruitment Result
- **Outputs:** Result Notification Message
- **Preconditions:** Candidate status has been updated to Selected, Rejected, or another outcome.
- **Postconditions:** Notification is delivered to the concerned student.

Acceptance Criteria:

1. Notification is generated when application results are published.
 2. Students receive notifications for status changes.
 3. Notification details accurately reflect the recruitment outcome.
 4. Notifications are accessible within the system.
- **Priority:** High
 - **Dependencies:** Job Application Management, Candidate Shortlisting

FR-9.4: Email Alerts

- **ID:** FR-9.4
- **Title:** Email Alerts
- **Description:** The system shall send email notifications for important placement activities, including announcements, interview schedules, application updates, and recruitment results.
- **Rationale:** Required to ensure users receive critical updates even when they are not actively using the portal.
- **Inputs:** Notification Content, Recipient Email Address

- **Outputs:** Email Alert Delivery Status
- **Preconditions:** Valid recipient email address exists, and email service is configured.
- **Postconditions:** Email notification is sent to the intended recipient.

Acceptance Criteria:

1. Email alerts are generated for configured events.
 2. Emails contain accurate and relevant information.
 3. Delivery failures are logged for administrative review.
 4. Users receive email notifications for important updates.
- **Priority:** High
 - **Dependencies:** Notification Management Module, Email Service Integration

FR-9.5: Notification Tracking and Delivery

- **ID:** FR-9.5
- **Title:** Notification Tracking and Delivery
- **Description:** The system shall maintain records of all notifications generated and delivered through the portal and email channels.
- **Rationale:** Required to ensure reliable communication and support auditing of notification activities.
- **Inputs:** Notification Event Data
- **Outputs:** Notification History and Delivery Status
- **Preconditions:** Notification event has been triggered.
- **Postconditions:** Notification records are stored and available for review.

Acceptance Criteria:

1. All notifications are logged in the system.
 2. Notification history is available for authorized users.
 3. Delivery status is recorded for each notification.
 4. Failed notifications can be identified and reviewed.
- **Priority:** Medium
 - **Dependencies:** Notification Management Module, Email Service Integration, Database System

Feature 10: Reporting and Analytics

Actors

- Super Admin
- Placement Admin

FR-10.1: Generate Placement Statistics

- **ID:** FR-10.1
- **Title:** Generate Placement Statistics
- **Description:** The system shall provide placement statistics, including the number of placed students, placement percentage, highest package, average package, and company participation details.
- **Rationale:** Required to evaluate placement performance and support strategic decision-making.
- **Inputs:** Placement Records, Academic Year, Department Filters
- **Outputs:** Placement Statistics Dashboard and Reports
- **Preconditions:** Placement data exists in the system.
- **Postconditions:** Placement statistics are generated and displayed to authorized users.

Acceptance Criteria:

1. Authorized users can view overall placement statistics.
2. Statistics are calculated accurately from placement records.
3. Filters can be applied to refine results.
4. Data is displayed in an understandable format.

- **Priority:** High
- **Dependencies:** Placement Records, Reporting Module

FR-10.2: Generate Student Reports

- **ID:** FR-10.2
- **Title:** Generate Student Reports
- **Description:** The system shall allow Super Admins and Placement Admins to generate reports containing student profile, application, and placement information.
- **Rationale:** Required for monitoring student participation and placement outcomes.
- **Inputs:** Student Filters, Report Parameters
- **Outputs:** Student Report
- **Preconditions:** Student records exist in the system.
- **Postconditions:** Student report is generated and available for viewing or export.

Acceptance Criteria:

1. Authorized users can generate reports for individual or multiple students.
2. Reports contain accurate student information.
3. Filters can be applied based on department, batch, and placement status.
4. Generated reports are available for export.

- **Priority:** High
- **Dependencies:** Student Profile Management, Reporting Module

FR-10.3: Generate Recruiter Reports

- **ID:** FR-10.3
- **Title:** Generate Recruiter Reports
- **Description:** The system shall allow authorized users to generate reports containing recruiter activities, job postings, hiring statistics, and recruitment outcomes.
- **Rationale:** Required to analyse recruiter engagement and hiring performance.
- **Inputs:** Recruiter Filters, Report Parameters
- **Outputs:** Recruiter Report
- **Preconditions:** Recruiter and recruitment data exist in the system.
- **Postconditions:** Recruiter report is generated and available for review.

Acceptance Criteria:

1. Authorized users can generate recruiter-specific reports.
2. Reports include recruitment and hiring statistics.
3. Data is accurate and up to date.
4. Reports can be exported.

- **Priority:** Medium
- **Dependencies:** Recruiter Management, Job Posting Management, Reporting Module

FR-10.4: Generate Department-wise Reports

- **ID:** FR-10.4
- **Title:** Generate Department-wise Reports
- **Description:** The system shall generate reports summarizing placement activities and outcomes for individual academic departments.
- **Rationale:** Required to evaluate departmental placement performance and identify trends.
- **Inputs:** Department Selection, Report Parameters
- **Outputs:** Department-wise Placement Report
- **Preconditions:** Department and placement data exist in the system.
- **Postconditions:** Department-wise report is generated successfully.

Acceptance Criteria:

1. Users can select a department for reporting.
 2. Report includes placement statistics and student participation details.
 3. Data is displayed accurately.
 4. Report can be exported for further analysis.
- **Priority:** High
 - **Dependencies:** Student Profile Management, Placement Records, Reporting Module

FR-10.5: Generate Year-wise Analysis

- **ID:** FR-10.5
- **Title:** Generate Year-wise Analysis
- **Description:** The system shall provide year-wise analysis of placement activities, recruitment trends, and placement outcomes.
- **Rationale:** Required to compare placement performance across different academic years.
- **Inputs:** Academic Year Selection, Analysis Parameters
- **Outputs:** Year-wise Placement Analysis Report
- **Preconditions:** Historical placement data exists in the system.
- **Postconditions:** Year-wise analysis is generated and displayed.

Acceptance Criteria:

1. Users can select one or more academic years for comparison.
 2. Analysis includes placement percentages, recruiter participation, and hiring trends.
 3. Results are generated accurately.
 4. Analysis can be exported.
- **Priority:** High
 - **Dependencies:** Placement Records, Reporting Module

FR-10.6: Export Reports

- **ID:** FR-10.6
- **Title:** Export Reports
- **Description:** The system shall allow authorized users to export generated reports in supported formats such as PDF and Excel.
- **Rationale:** Required to facilitate sharing, archiving, and offline analysis of placement data.
- **Inputs:** Report Selection, Export Format

- **Outputs:** Downloadable Report File
- **Preconditions:** Report has been successfully generated.
- **Postconditions:** Report is exported and made available for download.

Acceptance Criteria:

1. Users can export reports in supported formats.
2. Exported reports preserve report data and formatting.
3. Downloaded files can be opened without errors.
4. Export process completes successfully within acceptable response time.

- **Priority:** High
- **Dependencies:** Reporting Module, File Generation Service

FR-10.7: Dashboard Analytics

- **ID:** FR-10.7
- **Title:** Dashboard Analytics
- **Description:** The system shall provide a dashboard displaying key placement metrics, trends, and analytical summaries for Super Admins and Placement Admins.
- **Rationale:** Required to provide a quick overview of placement activities and system performance.
- **Inputs:** Placement Data, Recruitment Data, Student Data
- **Outputs:** Analytical Dashboard with Metrics and Charts
- **Preconditions:** Relevant placement and recruitment data exist in the system.
- **Postconditions:** Dashboard analytics are displayed to authorized users.

Acceptance Criteria:

1. Dashboard displays real-time placement statistics.
2. Key metrics are updated automatically.
3. Users can view summarized placement trends.
4. Dashboard data is consistent with generated reports.

- **Priority:** High
- **Dependencies:** Reporting Module, Placement Records, Recruitment Records, Student Database

7.0 Non-functional Requirements

7.1 Performance Requirements

NFR-1.1: Concurrent User Support

- **Description:** The system shall support at least 500 concurrent users without significant degradation in performance or availability.
- **Rationale:** To ensure the portal can handle peak placement season usage by students, recruiters, and administrators.
- **Metric:** Support ≥ 500 concurrent active users.
- **Verification:** Load testing using JMeter or equivalent performance testing tools.

NFR-1.2: Response Time

- **Description:** The system shall maintain an average response time of less than 3 seconds for all standard user operations under normal load conditions.
- **Rationale:** To provide a responsive and satisfactory user experience.
- **Metric:** Average response time < 3 seconds.
- **Verification:** Performance testing and monitoring under expected workload conditions.

NFR-1.3: Database Performance

- **Description:** The system shall optimize database operations through indexing, query optimization, and efficient data access mechanisms to ensure acceptable performance.
- **Rationale:** To minimize delays in data retrieval and processing for placement-related activities.
- **Metric:** Database queries for common operations shall complete within 1 second under normal load.
- **Verification:** Database performance testing, query analysis, and benchmarking.

7.2 Safety Requirements

NFR-2.1: Database Backup

- **Description:** The system shall perform periodic database backups to ensure that critical placement and user data can be recovered in the event of system failure or data corruption.
- **Rationale:** To protect organizational data and minimize the risk of permanent data loss.

- **Metric:** Automated backup executed at least once every 24 hours.
- **Verification:** Backup schedule audits and restoration testing.

NFR-2.2: Disaster Recovery

- **Description:** The system shall provide disaster recovery procedures to restore system operations and data following hardware failures, software failures, or other catastrophic events.
- **Rationale:** To ensure business continuity and minimize service disruption.
- **Metric:** Recovery Time Objective (RTO) ≤ 4 hours and Recovery Point Objective (RPO) ≤ 24 hours.
- **Verification:** Periodic disaster recovery drills and recovery testing.

NFR-2.3: Data Loss Prevention

- **Description:** The system shall implement mechanisms to prevent accidental or unauthorized loss of data through backup, recovery, validation, and access control measures.
- **Rationale:** To maintain data integrity and reliability of placement-related information.
- **Metric:** No critical data loss during normal system operations and recovery scenarios.
- **Verification:** Data integrity testing, backup validation, and security audits.

7.3 Security Requirements

NFR-3.1: JWT-Based Authentication

- **Description:** The system shall use JSON Web Tokens (JWT) for user authentication and authorization of protected resources.
- **Rationale:** To provide secure, stateless authentication for web application users.
- **Metric:** 100% of protected API requests require a valid JWT token.
- **Verification:** Security testing and API authentication validation.

NFR-3.2: Password Encryption

- **Description:** The system shall store user passwords in encrypted form using industry-standard hashing algorithms.
- **Rationale:** To protect user credentials from unauthorized access and data breaches.
- **Metric:** 100% of stored passwords shall be hashed and salted.
- **Verification:** Security audit and database inspection.

NFR-3.3: Role-Based Access Control (RBAC)

- **Description:** The system shall enforce Role-Based Access Control to ensure users can access only the functionalities permitted for their assigned roles.
- **Rationale:** To protect sensitive information and prevent unauthorized actions.
- **Metric:** 100% of restricted resources shall require role validation before access.
- **Verification:** Authorization testing and security reviews.

NFR-3.4: HTTPS Communication

- **Description:** The system shall use HTTPS for all client-server communication to ensure secure data transmission.
- **Rationale:** To protect sensitive information from interception and tampering during transmission.
- **Metric:** 100% of network communication shall occur over HTTPS.
- **Verification:** SSL/TLS configuration testing and security scans.

NFR-3.5: Input Validation

- **Description:** The system shall validate all user inputs on both client-side and server-side before processing.
- **Rationale:** To prevent invalid data entry and reduce security vulnerabilities.
- **Metric:** 100% of input fields shall have validation rules applied.
- **Verification:** Functional testing and security testing.

NFR-3.6: SQL Injection Prevention

- **Description:** The system shall prevent SQL injection attacks by using parameterized queries, prepared statements, and ORM-based data access mechanisms.
- **Rationale:** To protect the database from unauthorized access and manipulation.
- **Metric:** Zero successful SQL injection vulnerabilities identified during security testing.
- **Verification:** Penetration testing and code review.

NFR-3.7: Secure Session Management

- **Description:** The system shall securely manage user sessions through token expiration, session invalidation on logout, and protection against session hijacking.
- **Rationale:** To maintain secure authenticated user sessions.
- **Metric:** All sessions shall expire after a configurable inactivity period and be invalidated upon logout.
- **Verification:** Security testing and session management validation.

7.4 Software Quality Attributes

NFR-4.1: Reliability

- **Description:** The system shall operate reliably and maintain an uptime of at least 99% during normal operational periods.
- **Rationale:** To ensure continuous access to placement services and minimize disruptions.
- **Metric:** System availability $\geq 99\%$ per month.
- **Verification:** System monitoring reports and uptime analysis.

NFR-4.2: Availability

- **Description:** The placement portal shall be accessible to authorized users 24 hours a day, 7 days a week, except during scheduled maintenance periods.
- **Rationale:** To ensure users can access the system whenever required.
- **Metric:** Portal accessibility available 24×7 with scheduled maintenance notifications.
- **Verification:** Availability monitoring and operational logs.

NFR-4.3: Maintainability

- **Description:** The system shall follow a modular and layered architecture to facilitate maintenance, enhancements, and bug fixes.
- **Rationale:** To reduce maintenance effort and support future development activities.
- **Metric:** System modules shall be independently maintainable and follow established coding standards.
- **Verification:** Architecture review, code review, and maintainability assessment.

NFR-4.4: Scalability

- **Description:** The system shall support growth in the number of users, recruiters, placement drives, and institutional data without significant degradation in performance.
- **Rationale:** To accommodate future expansion of placement activities and user base.
- **Metric:** System shall support at least a 50% increase in workload without major architectural changes.
- **Verification:** Scalability testing and capacity planning analysis.

NFR-4.5: Usability

- **Description:** The system shall provide an intuitive, user-friendly, and consistent interface that enables users to perform tasks with minimal training.
- **Rationale:** To improve user satisfaction and reduce operational errors.

- **Metric:** Users shall be able to complete common tasks with minimal assistance and achieve at least an 85% task success rate during usability testing.
- **Verification:** Usability testing, user feedback, and user acceptance testing.

7.5 Other Requirements

NFR-5.1: Mobile Responsive Design

- **Description:** The system shall provide a responsive user interface that adapts to different screen sizes and devices, including desktops, tablets, and smartphones.
- **Rationale:** To ensure users can access the placement portal effectively from various devices.
- **Metric:** All major application pages shall render correctly on-screen widths ranging from 320px to 1920px.
- **Verification:** Responsive design testing across multiple devices and screen resolutions.

NFR-5.2: Multi-Browser Compatibility

- **Description:** The system shall function correctly across major modern web browsers.
- **Rationale:** To provide a consistent user experience regardless of the browser used.
- **Metric:** Full compatibility with the latest two versions of Chrome, Firefox, Edge, and Safari.
- **Verification:** Cross-browser testing and user acceptance testing.

NFR-5.3: Administrative Audit Logs

- **Description:** The system shall maintain audit logs of critical administrative activities, including user management, account modifications, job postings, and system configuration changes.
- **Rationale:** To support accountability, monitoring, troubleshooting, and compliance requirements.
- **Metric:** 100% of critical administrative actions shall be logged with user ID, timestamp, and activity details.
- **Verification:** Audit log review and system testing.

NFR-5.4: Exportable Reports

- **Description:** The system shall support exporting generated reports in commonly used formats such as PDF and Excel.
- **Rationale:** To facilitate reporting, record keeping, and data sharing among stakeholders.
- **Metric:** All standard reports shall be exportable in at least PDF and XLSX formats.
- **Verification:** Functional testing of report export features.

NFR-5.5: Centralized Error Logging

- **Description:** The system shall maintain centralized logs for application errors, exceptions, and system events.
- **Rationale:** To simplify troubleshooting, monitoring, and maintenance activities.
- **Metric:** 100% of application exceptions and critical system errors shall be recorded in centralized logs.
- **Verification:** Error simulation testing and log review.

7.6 Applicable Standards

NFR-6.1: REST API Compliance

- **Description:** The system shall implement RESTful API principles for all backend service endpoints, including the use of standard HTTP methods, status codes, and resource-based URL structures.
- **Rationale:** To ensure consistency, interoperability, and maintainability of system integrations.
- **Metric:** 100% of public APIs shall conform to REST architectural principles.
- **Verification:** API design review, endpoint testing, and architecture validation.

NFR-6.2: OWASP Security Compliance

- **Description:** The system shall adhere to OWASP security guidelines and best practices to protect against common web application vulnerabilities.
- **Rationale:** To enhance application security and reduce exposure to cyber threats.
- **Metric:** No critical vulnerabilities identified from the OWASP Top 10 security categories during security assessment.
- **Verification:** Security audits, vulnerability scanning, and penetration testing.

NFR-6.3: JWT Authentication Compliance

- **Description:** The system shall implement authentication and authorization mechanisms in accordance with JWT standards for secure token generation, validation, and expiration management.
- **Rationale:** To ensure secure and standardized user authentication across the application.
- **Metric:** 100% of authenticated API requests shall use valid JWT tokens.
- **Verification:** Authentication testing, token validation testing, and security review.

NFR-6.4: SQL Standards Compliance

- **Description:** The system shall use standardized SQL practices and database design principles to ensure data integrity, portability, and maintainability.
- **Rationale:** To support efficient database operations and compliance with established database development standards.
- **Metric:** Database schema and queries shall conform to SQL standards and normalization principles where applicable.
- **Verification:** Database design review, query analysis, and code inspection.

8.0 Acceptance Criteria

The system shall be accepted when:

1. Super Admin can manage Placement Admin accounts successfully.
2. Placement Admin can create and manage student accounts.
3. Students can login using assigned credentials.
4. Students can update profiles and upload resumes.
5. Recruiters can post job opportunities.
6. Placement Admin can create placement drives.
7. Eligibility verification functions correctly.
8. Students can apply for eligible drives.
9. Recruiters can shortlist candidates.
10. Application statuses are updated correctly.
11. Notifications are delivered successfully.
12. Reports and analytics are generated accurately.
13. Authentication and authorization function correctly.
14. All critical test cases pass successfully.

9.0 Deliverables

1. Software Requirements Specification (SRS)
2. System Design Document (SDD)
3. Source Code Repository
4. React Frontend Application
5. Spring Boot Backend Application
6. MySQL Database Scripts
7. API Documentation
8. User Manual
9. Installation Guide
10. Deployment Guide
11. Test Cases
12. Test Execution Report
13. Final Project Report
14. Project Presentation
15. System Demonstration